GOVIND RANADAS

ELECTRO-OPTICS ENGINEER

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DOB: 18/05/1992 **Passport:** R8040933

CAREER OBJECTIVE

Highly motivated Development Engineer with extensive experience in the design, development, testing, qualification, and cross-functional project execution of opto-electronic and photonic systems. Possesses strong hands-on expertise in optical metrology, fiber optics, laser testing, PCB layout, and advanced simulation & design tools (Altium, LTSpice, Zemax, Lumerical). Seeking a challenging role in a forward-looking organization where I can leverage analytical thinking, hands-on engineering skills, and system integration capabilities to drive operational excellence.

WORK EXPERIENCE

ENGINEER - 1 (ELECTRO-OPTICS) | Halcon Systems LLC (Abu Dhabi, UAE) — April 2025 – Present

- **Camera & Sensor Testing:** Supported the development team with testing of different cameras and sensors, capturing output data for seeker development.
- **Fiber-Optic Assembly & Testing:** Executed precision fiber splicing, cleaving, and optical loss measurements for fiber-optic systems.
- **PCB Design & Layout Development:** Designed and developed PCB layouts for electronic circuits, including component placement, routing, and design modifications.
- **Schematic Development & ICD Analysis:** Reviewed and interpreted Interface Control Documents (ICDs) to understand electrical interfaces, signal requirements, pin configurations, and system integration specifications to develop circuit schematics accordingly.
- **ICD Documentation & Maintenance:** Prepared and maintained Interface Control Documents (ICDs), documenting electrical interfaces, connector pinouts, signal specifications, and interconnection requirements.

OPTO-ELECTRONICS ENGINEER | Halcon Systems LLC (Abu Dhabi, UAE) — April 2023 – March 2025

- **Seeker System Testing & Qualification:** Executed comprehensive testing and qualification of seeker systems used in guided weapon platforms to ensure accuracy, reliability, and compliance with defense standards.
- **On-Field Weapon Trial Support:** Supported live and on-field weapon trials by providing hands-on technical expertise, system validation, and cross-team coordination.
- **Laboratory Instrument Calibration & Validation:** Supported the calibration and validation of laboratory instruments, ensuring accurate, reliable, and repeatable data in compliance with established laboratory standards.
- **Technical Evaluation for Procurement:** Provided detailed technical evaluations, feasibility assessments, and performance insights to support strategic procurement decisions.
- **Technical Reporting & Documentation:** Prepared structured technical reports and documentation to communicate test outcomes, risks, and recommendations to engineering and procurement stakeholders.

GRADUATE ENGINEER (PHOTONICS / OPTICS) | Compass Tech Enterprises — January 2018 – December 2021

(Thrissur, India)

- **Optical & Photonics Engineering Development:** Investigated, developed, and implemented optical and photonics engineering solutions to support product development and manufacturing processes.
- **Optical System Design & Analysis:** Designed, modeled, and analyzed optical components and subsystems to reliably achieve defined functional and performance requirements.
- **Project Cost & Schedule Support:** Collaborated with cross-functional teams to ensure engineering deliverables aligned with project cost controls and timing constraints.
- **Technical Documentation & Presentation:** Prepared thorough technical documentation and professional presentations detailing design approaches, test results, and engineering outcomes.

EDUCATION

5-Year Integrated Master of Science (MSc) in Photonics (CGPA: 6.44) — **2012 – 2017**
International School of Photonics, Cochin University of Science and Technology, Kerala, India

ACADEMIC PROJECTS

Master's Project: Diode-Pumped Solid-State Laser (DPSSL) **Dec 2016 – May 2017**
Holmarc Opto-Mechatronics Ltd, Cochin, India

- Developed a functional diode-pumped solid-state laser by applying solid-state laser optics, diode pumping principles, and optical alignment techniques.
- Conducted system assembly, alignment, and performance evaluation to achieve stable and reliable laser output.
- Prepared detailed technical documentation and presented the laser model and results to academic and industry reviewers.

Bachelor's Project: Wavelength Division Multiplexing and De-Multiplexing Using Prism and Grating **Dec 2014 – May 2015**
International School of Photonics, CUSAT, India

- Applied core fiber-optic communication concepts to design and implement a WDM and de-multiplexing system using prisms and diffraction gratings.
- Designed precise optical alignment techniques to enable effective wavelength separation and signal integrity.
- Documented the complete project methodology and successfully presented the working model and outcomes.

SKILLS

Software Skills:	Altium | LTSpice | Optiwave Design Software | OptiSystem | Zemax | Lumerical FDTD | Python | MS Office Suite (Word, Excel, PowerPoint)
Professional Skills:	Optical System Design & Analysis | Opto-Electronics & Photonics Engineering | PCB Design & Circuit Simulation | Optical Testing & Qualification | System Integration | Configuration Management | Technical Documentation & Reporting | On-Field Testing & Technical Support | Problem Solving
Personal Skills:	Analytical Thinking | Attention to Detail | Time Management | Adaptability | Teamwork | Communication Skills | Self-Motivation | Continuous Learning | Work Ethic

LANGUAGES

Mother tongue(s): Malayalam | **Other language(s):** ENGLISH (Listening C2, Reading C2, Writing B2, Spoken Production B2, Spoken Interaction B2), HINDI